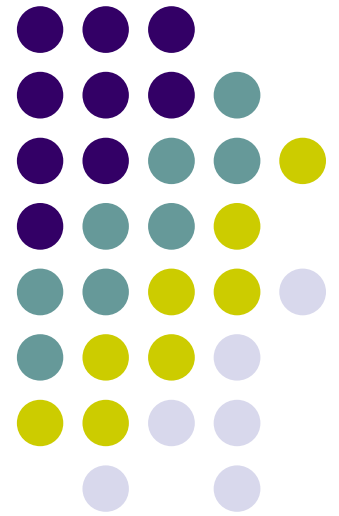


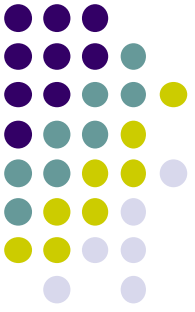
WIA1005 Network Technology Foundation

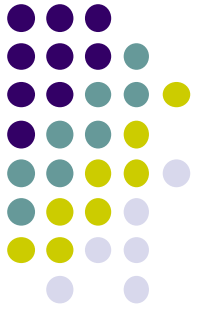
Chapter 5 IPv4



Contents

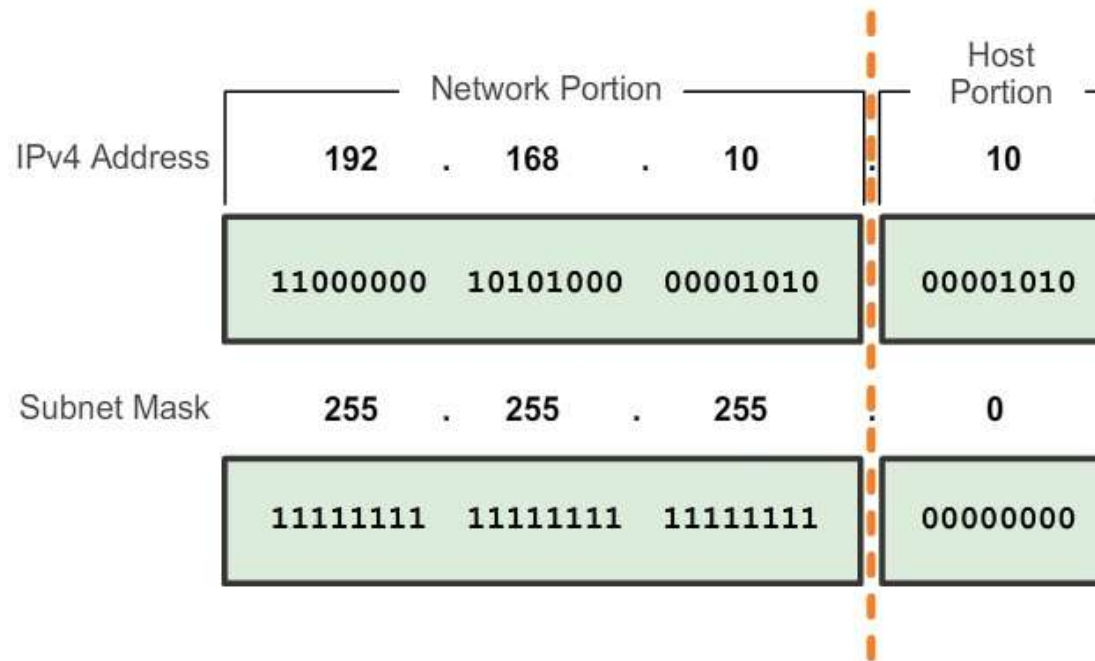
- IPv4
- Subnetting

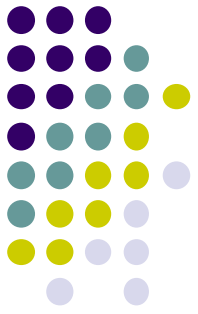




IPv4

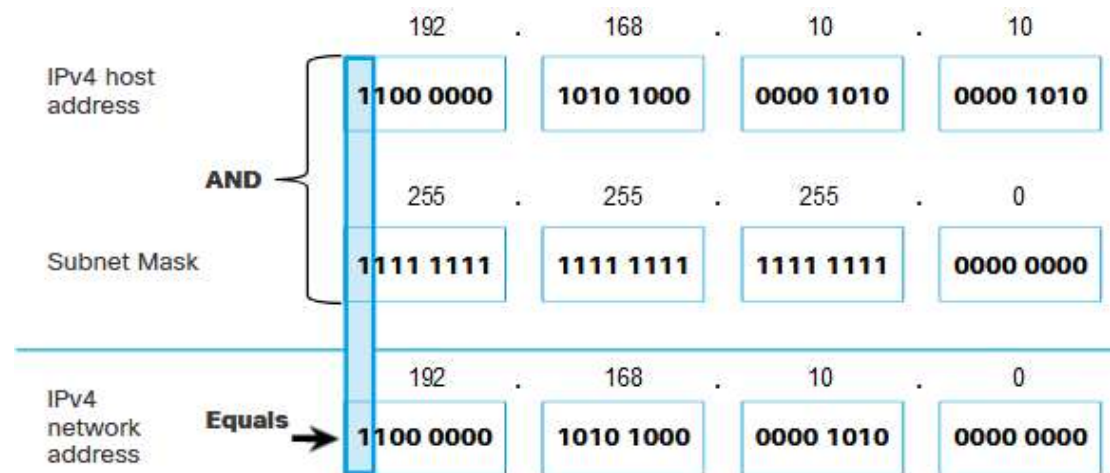
- IPv4 consists of 32-bits. A.B.C.D (0-255)
- Within the 32-bit, a portion of the bits makes up the **network** and a portion of the bits makes up the **host**.



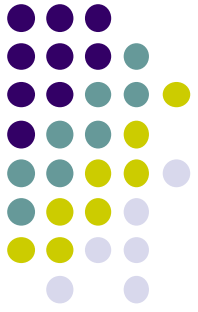


IPv4

- IPv4 subnet mask is used to differentiate the network portion from the host portion of an IPv4 address.
- When an IPv4 address is assigned to a device, the subnet mask is used to determine the network address of the device.
- The actual process used to identify the network portion and host portion is called ANDing.

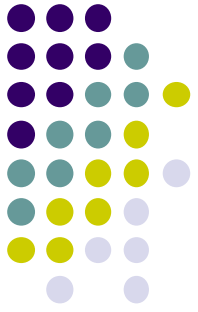


IPv4



- Within each network are three types of IP addresses:
 - Network address
 - Host addresses
 - Broadcast address
- **Network Address**
 - The network address is a standard way to refer to a network. It has the same subnet mask (network bit).
 - It is located on the same broadcast domain as other hosts with the same network address.

IPv4

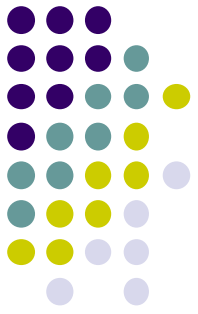


- **Host Address**

- The unique address that is assigned to the end device. In IPv4, the host address is the addresses in between the network address and the broadcast address. (first host and last host)

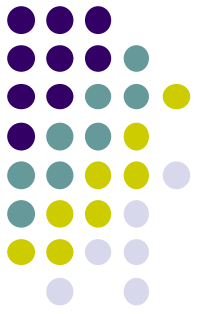
- **Broadcast Address**

- The IPv4 broadcast address is a special address for each network that allows communication to all the hosts in that network.
- The broadcast address uses the highest address in the network range. This is the address in which the bits in the host portion are all 1s.



IPv4

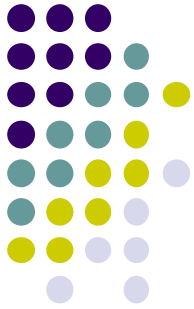
- IPv4 host addresses are unicast addresses and are in the address range of 0.0.0.0 to 223.255.255.255.
- IPv4 has a block of addresses reserved for addressing **multicast** groups. This address range is **224.0.0.0 to 239.255.255.255**.
- The multicast address range is subdivided into different types of addresses: 224.0.0.0 to 224.0.0.255 are reserved link local addresses, and the globally scoped addresses are 224.0.1.0 to 238.255.255.255.



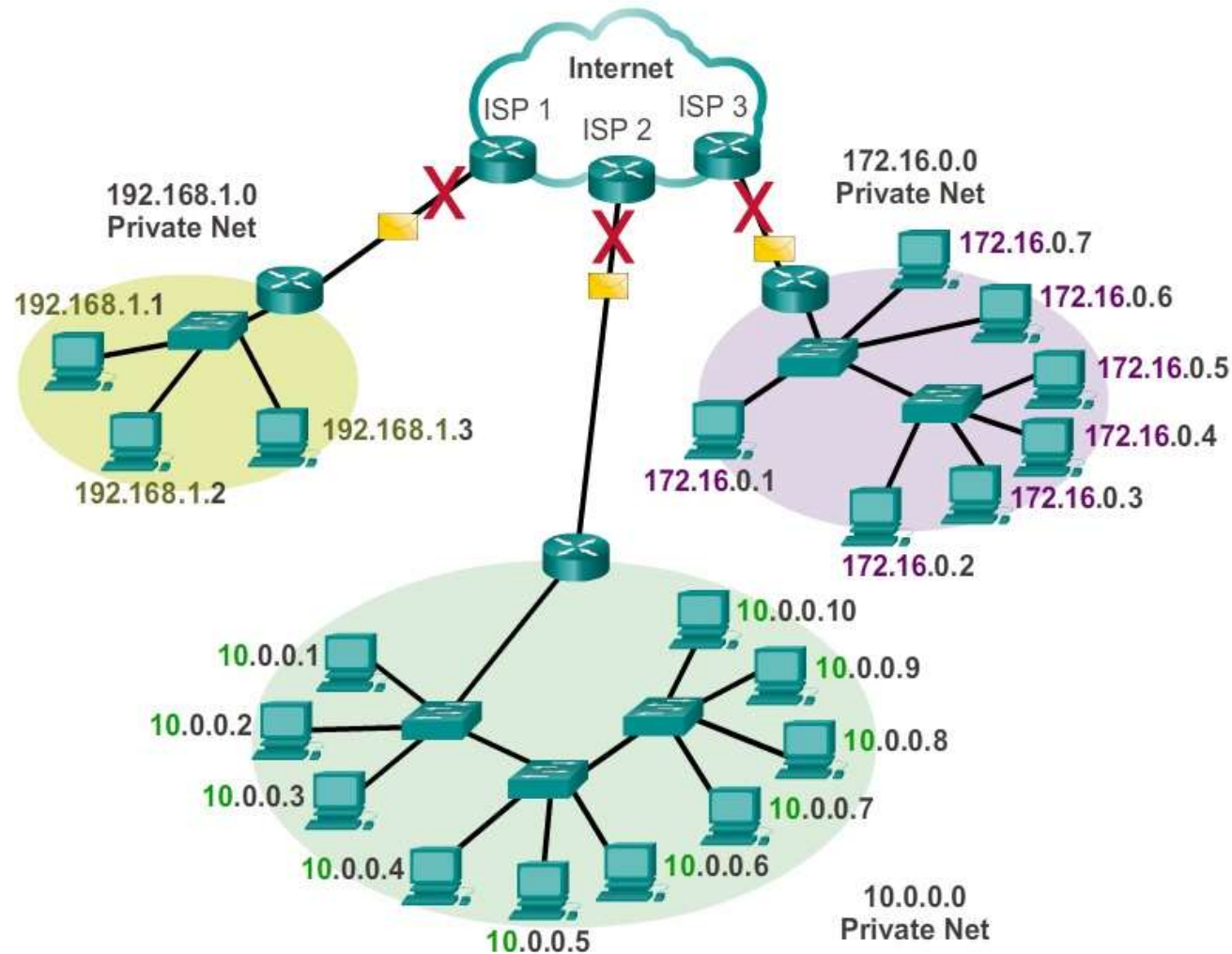
IPv4

- **Private addresses** are defined in RFC 1918 that are used in private networks. Hosts that do not require access to the Internet can use private addresses. However within the private network, hosts still require unique IP addresses within the private space.
- The private address blocks are:
 - **10.0.0.0 to 10.255.255.255 (10.0.0.0/8)**
 - **172.16.0.0 to 172.31.255.255 (172.16.0.0/12)**
 - **192.168.0.0 to 192.168.255.255 (192.168.0.0/16)**
- The vast majority of the addresses in the IPv4 unicast host range are **public addresses**. These addresses are designed to be used in the hosts that are publicly accessible from the Internet.

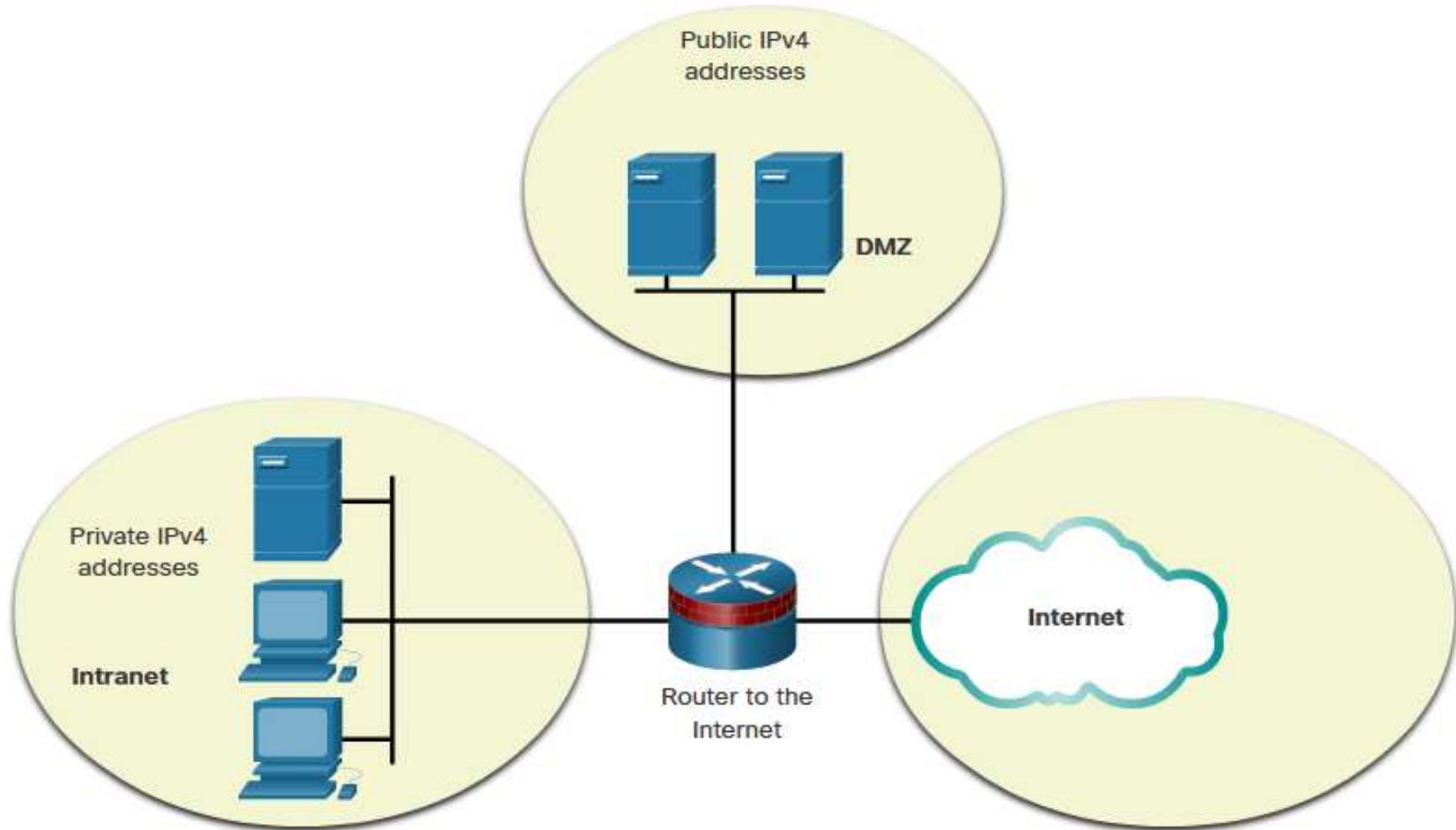
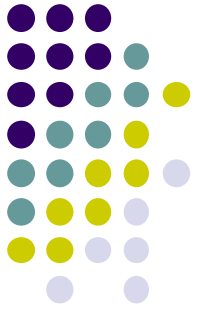
IPv4

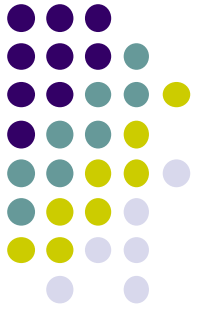


Private addresses cannot be routed over the Internet



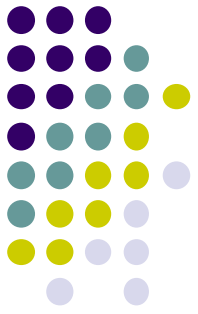
IPv4





IPv4

- **IPv4 loopback address** 127.0.0.1 is a special address that hosts use to direct traffic to themselves. The loopback address creates a shortcut method for TCP/IP applications and services that run on the same device to communicate with one another.
- IPv4 addresses in the address block **169.254.0.0 to 169.254.255.255 (169.254.0.0/16)** are designated as link-local addresses. These addresses can be automatically assigned to the local host by the operating system in environments where **no IP configuration is available**.

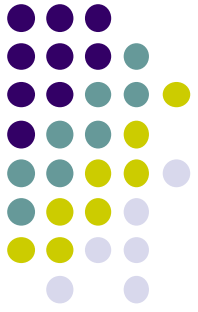


IPv4

- Classful addressing grouped the unicast ranges into specific sizes called class A, class B, and class C addresses. It also defined class D (multicast) and class E (experimental).

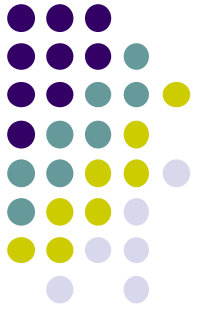
IP Address Classes				
Address Class	1st octet range (decimal)	1st octet bits (green bits do not change)	Network (N) and Host (H) parts of address	Default subnet mask (decimal and binary)
A	1-127**	00000000-01111111	N.H.H.H	255.0.0.0
B	128-191	10000000-10111111	N.N.H.H	255.255.0.0
C	192-223	11000000-11011111	N.N.N.H	255.255.255.0
D	224-239	11100000-11101111	NA (multicast)	
E	240-255	11110000-11111111	NA (experimental)	

IPv4



Assignment of IP Addresses



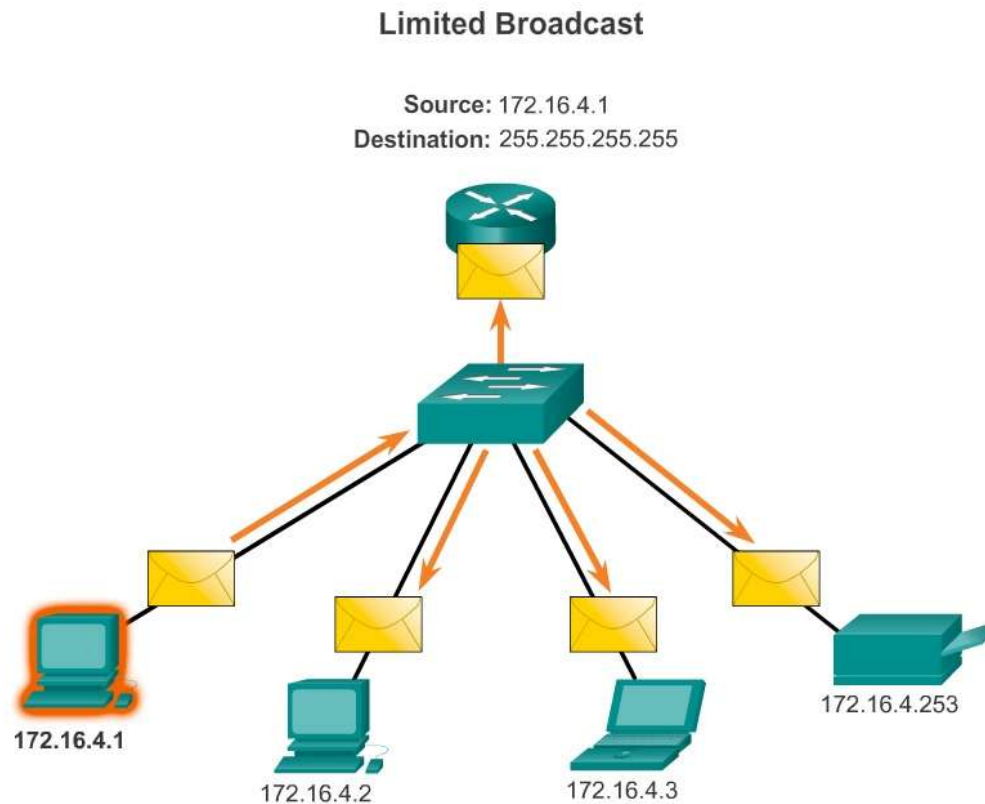


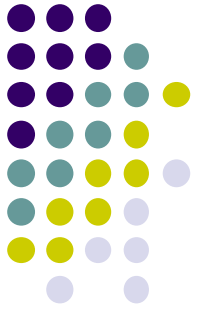
Subnetting

- A large broadcast domain is a network that connects many hosts.
- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network. This results in slow network operations due to the significant amount of traffic it can cause, and slow device operations because a device must accept and process each broadcast packet.
- The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.
- These smaller network spaces are called subnets.

Subnetting

- Subnetting can reduce overall network traffic and improve network performance.

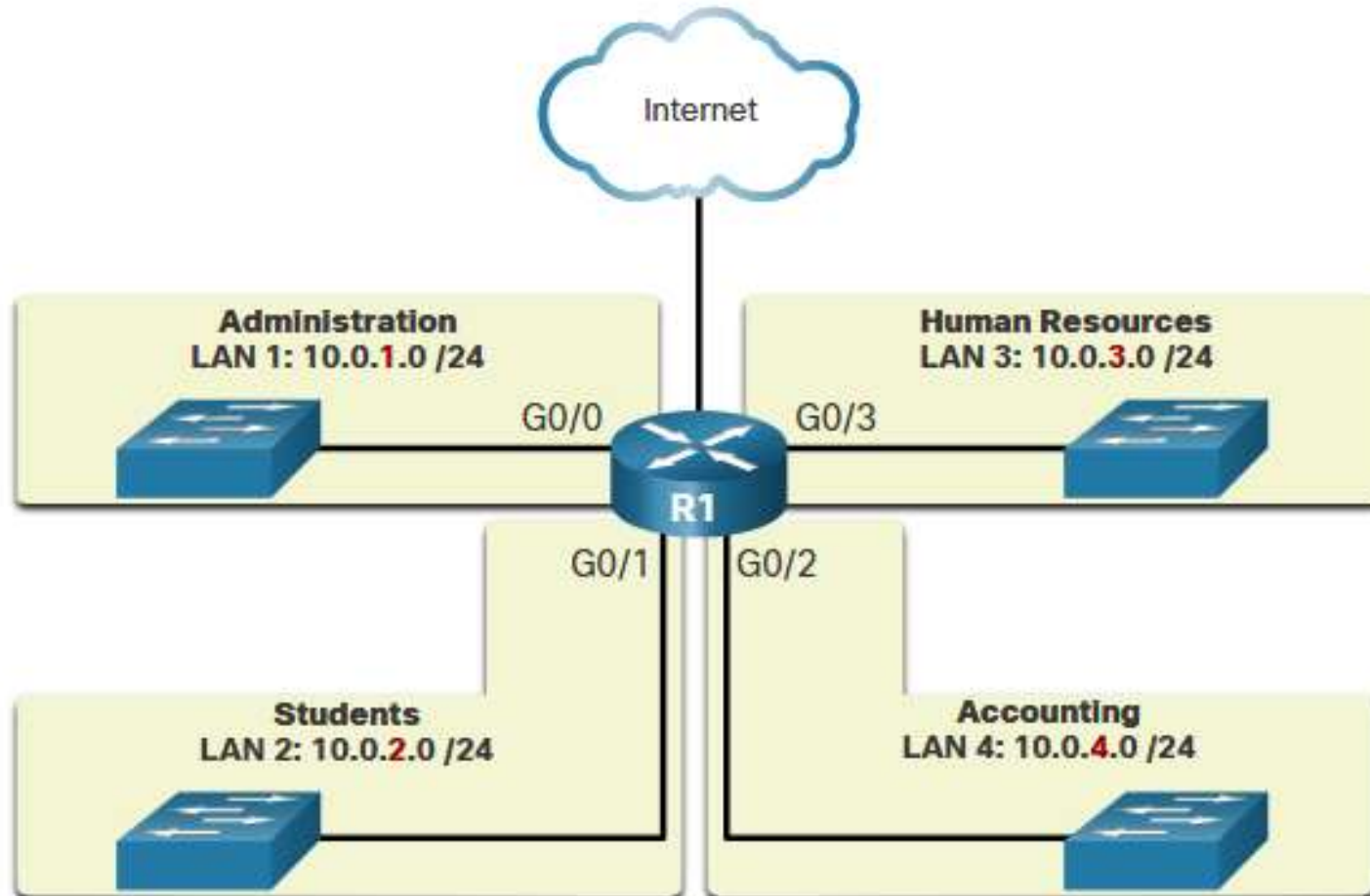
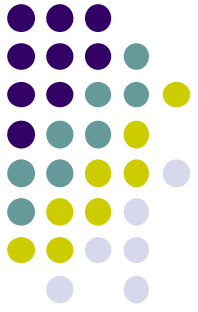


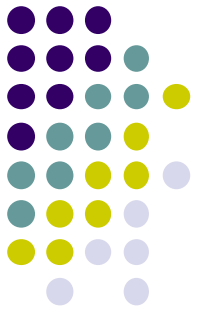


Subnetting

- A router is necessary for devices on different subnets to communicate. Devices on a subnet use the router interface attached to their LAN as their default gateway.
- Traffic cannot be forwarded between subnets without the use of a router. Every interface on the router must have an IPv4 host address that belongs to the network or subnet to which the router interface is connected.
- Planning network subnets requires examination of both the needs of an organization's network usage, and how the subnets will be structured.

Subnetting

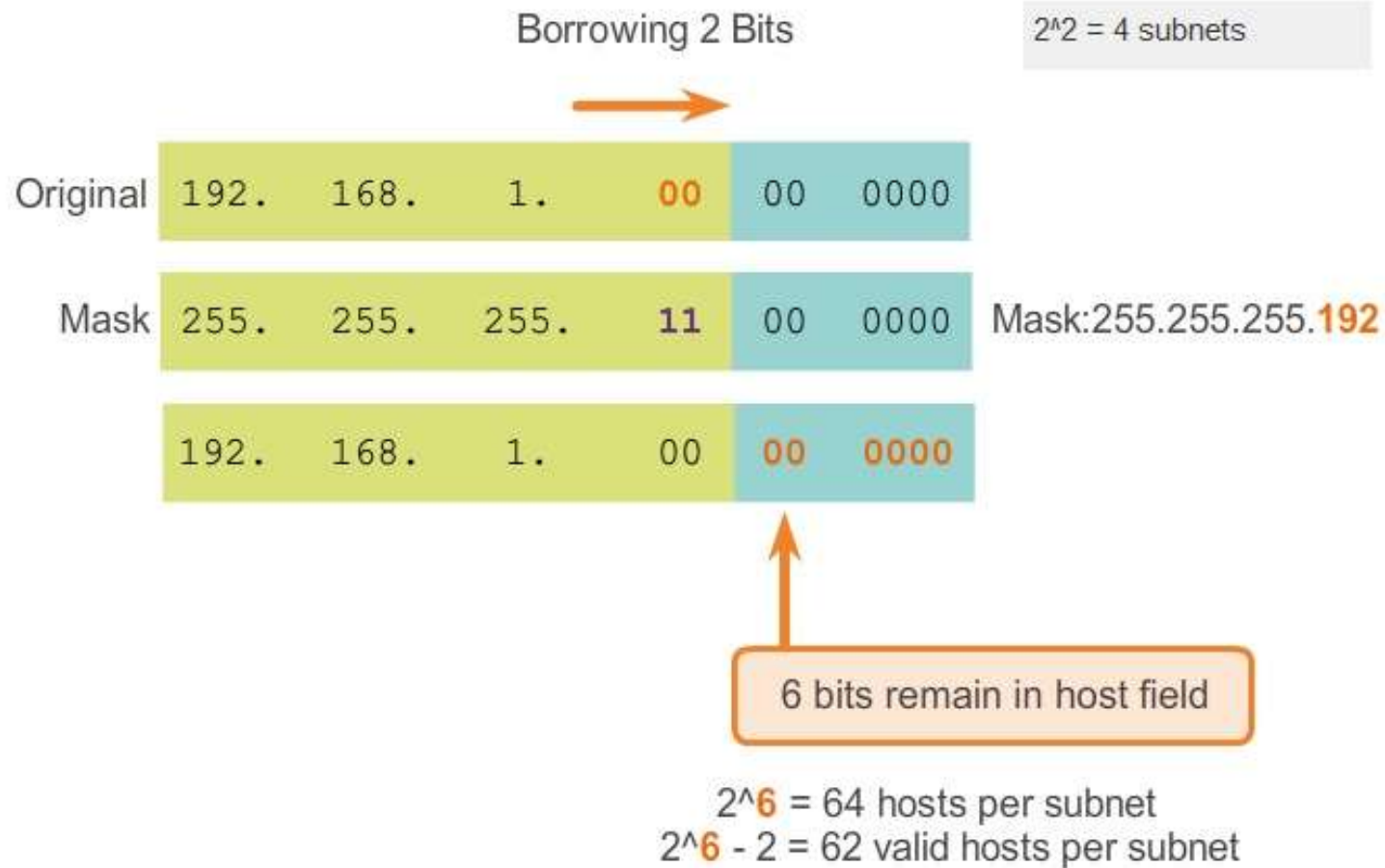
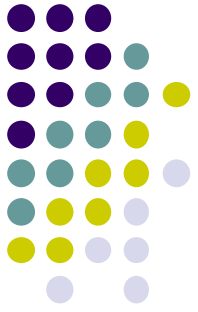




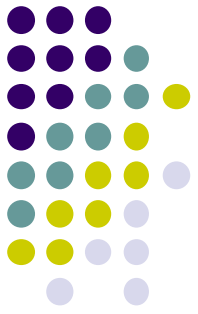
Subnetting

- The address plan includes deciding the needs for each subnet in terms of size, how many hosts per subnet, how host addresses will be assigned, which hosts will require static IP addresses and which hosts can use DHCP for obtaining their addressing information.
- Two very important factors that will lead to the determination of which private address block is required, are the **number of subnets required** and the **maximum number of hosts needed per subnet**.
- **Number of subnet = $2^{\text{borrowbit}}$**
- **Number of usable host = $2^{\text{hostbit}} - 2$**

Subnetting



Subnetting



Address Range for 192.168.1.0/26 Subnet

Network Address

192.	168.	1.	00	00	0000
------	------	----	----	----	------

 = 192.168.1.0

First Host Address

192.	168.	1.	00	00	0001
------	------	----	----	----	------

 = 192.168.1.1

Last Host Address

192.	168.	1.	00	11	1110
------	------	----	----	----	------

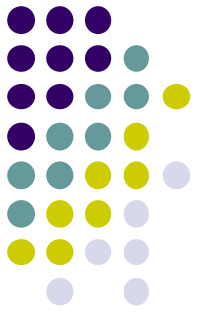
 = 192.168.1.62

Broadcast Address

192.	168.	1.	00	11	1111
------	------	----	----	----	------

 = 192.168.1.63

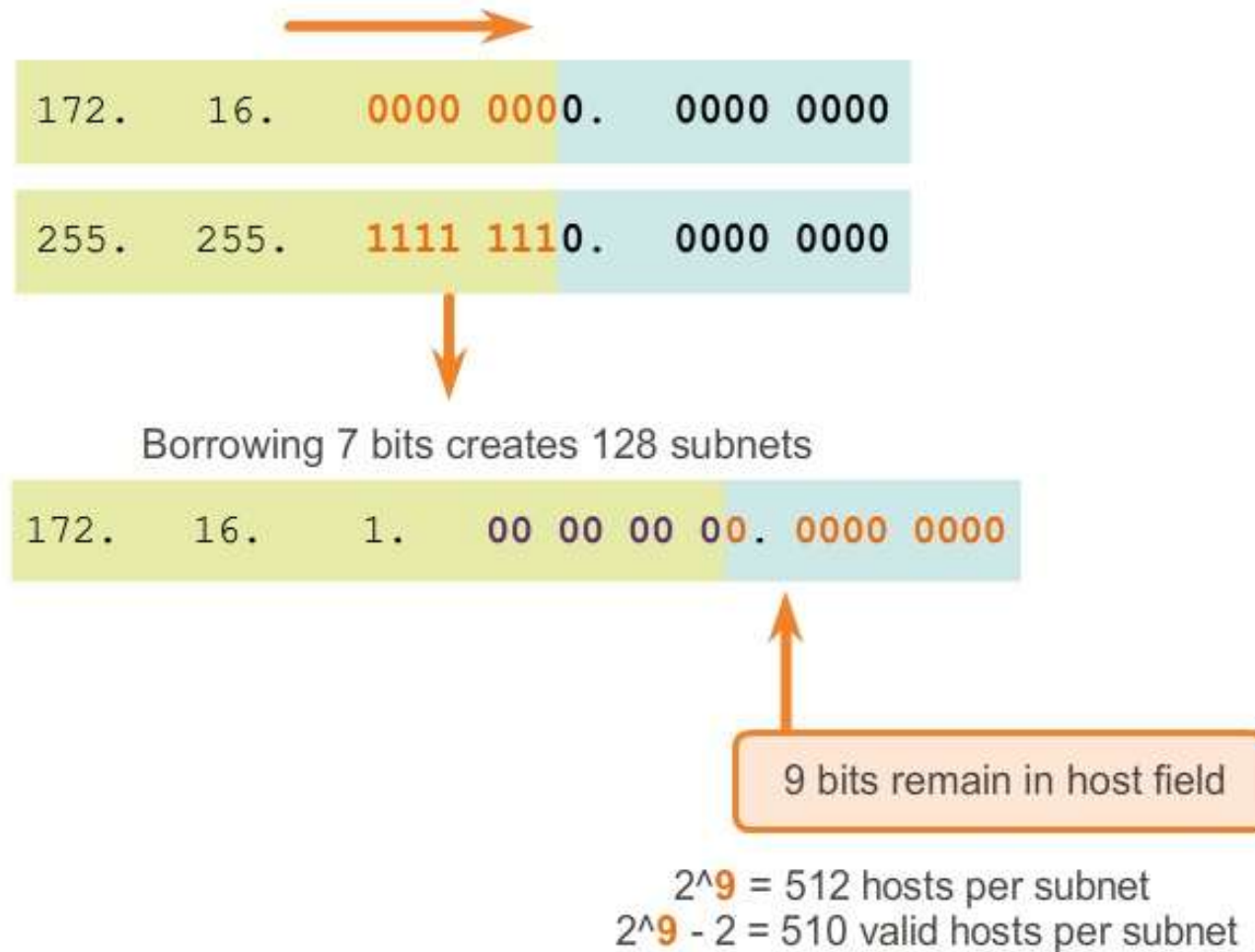
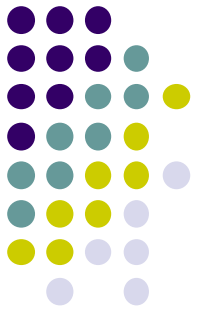
Subnetting



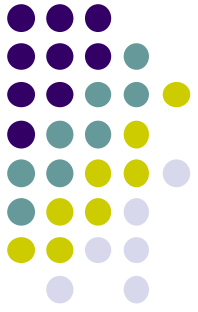
Subnet a /24 Network

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nhhhhhhh 11111111.11111111.11111111.10000000	2	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnhhhhhh 11111111.11111111.11111111.11000000	4	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnhhhhh 11111111.11111111.11111111.11100000	8	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnhhhh 11111111.11111111.11111111.11110000	16	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnhhh 11111111.11111111.11111111.11111000	32	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnnhh 11111111.11111111.11111111.11111100	64	2

Subnetting



Subnetting



Address Range for 172.16.0.0/23 Subnet

Network Address

172. 16. 1. 00 00 00 00. 0000 0000 = 172.16.0.0/23

First Host Address

172. 16. 1. 00 00 00 00. 0000 0001 = 172.16.0.1/23

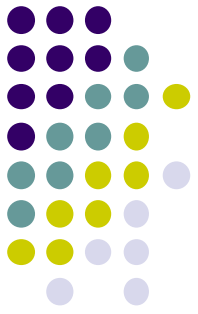
Last Host Address

172. 16. 1. 00 00 00 01. 1111 1110 = 172.16.0.254/23

Broadcast Address

172. 16. 1. 00 00 00 01. 1111 1111 = 172.16.1.255

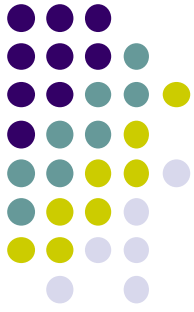
Subnetting



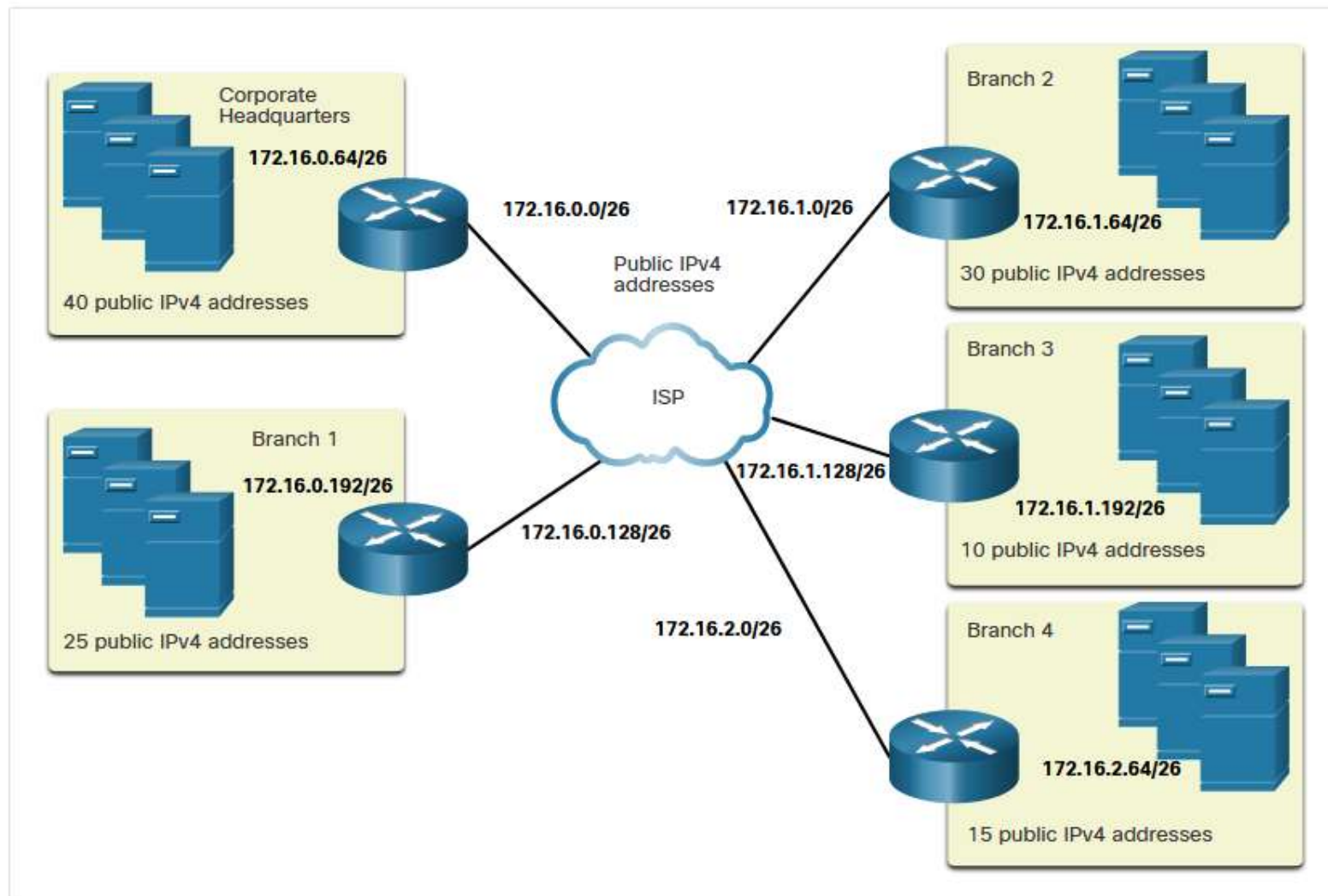
Subnet a /16 Network

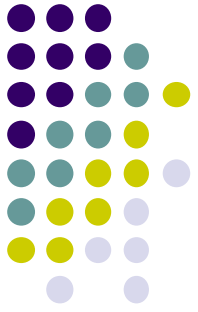
Prefix Length	Subnet Mask	Network Address (n = network, h = host)	# of subnets	# of hosts
/17	255.255.128.0	nnnnnnnn.nnnnnnnn.nhhhhhhh.hhhhhhhh 11111111.11111111.10000000.00000000	2	32766
/18	255.255.192.0	nnnnnnnn.nnnnnnnn.nnhhhhhh.hhhhhhhh 11111111.11111111.11000000.00000000	4	16382
/19	255.255.224.0	nnnnnnnn.nnnnnnnn.nnnhhhhh.hhhhhhhh 11111111.11111111.11100000.00000000	8	8190
/20	255.255.240.0	nnnnnnnn.nnnnnnnn.nnnnhhhh.hhhhhhhh 11111111.11111111.11110000.00000000	16	4094
/21	255.255.248.0	nnnnnnnn.nnnnnnnn.nnnnnhhh.hhhhhhhh 11111111.11111111.11111000.00000000	32	2046
/22	255.255.252.0	nnnnnnnn.nnnnnnnn.nnnnnnhh.hhhhhhhh 11111111.11111111.11111100.00000000	64	1022
/23	255.255.254.0	nnnnnnnn.nnnnnnnn.nnnnnnnh.hhhhhhhh 11111111.11111111.11111110.00000000	128	510
/24	255.255.255.0	nnnnnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh 11111111.11111111.11111111.00000000	256	254

Subnetting



Subnet Assignments to each Site and ISP



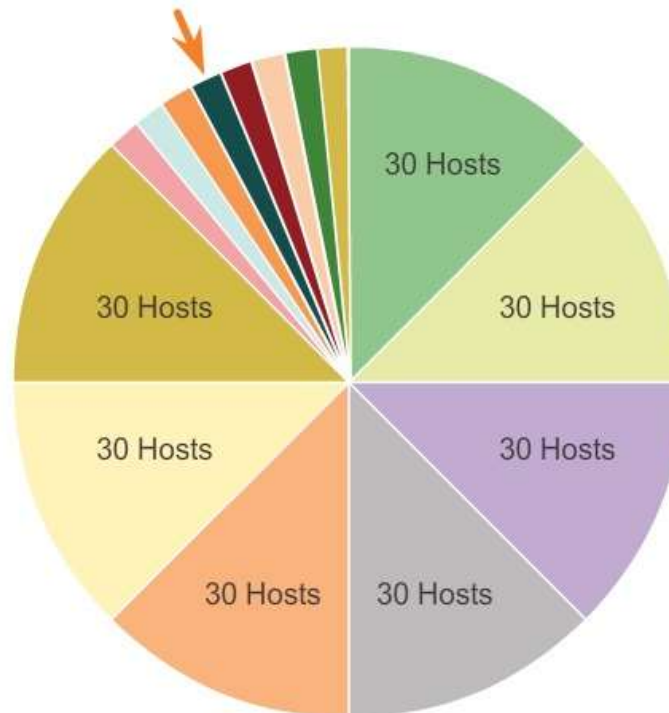


Subnetting

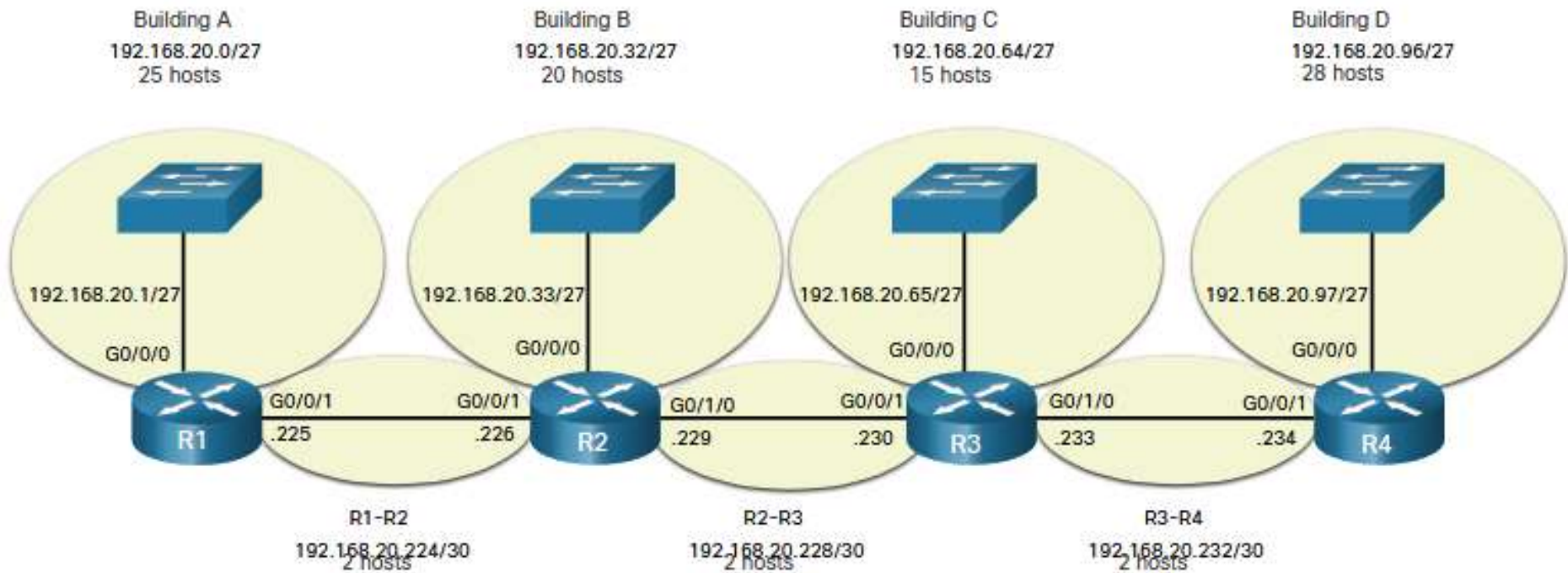
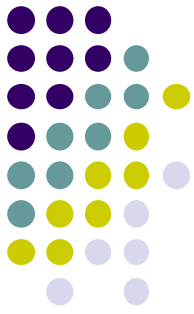
- Subnetting a subnet, or using **Variable Length Subnet Mask (VLSM)**, was designed to avoid wasting addresses.

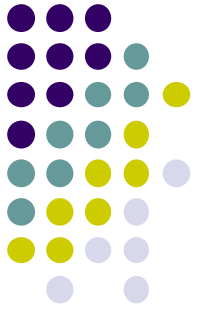
Subnets of Varying Sizes

One subnet was further divided to create 8 smaller subnets of 4 hosts each



Subnetting





Subnetting

- There are three primary considerations when planning address allocation.
 - **Preventing Duplication of Addresses** - Each host in an internetwork must have a unique address.
 - **Providing and Controlling Access** - Some hosts, such as servers, provide resources to internal hosts as well as to external hosts. Servers and peripherals should have a predictable **static IP** address.
 - **Monitoring Security and Performance** - Similarly, the security and performance of network hosts and the network as a whole must be monitored.
(Intermediary devices)